computing resources through source code, which can be treated just like any software system.

Using tools like Ansible, we can specify how a server or a network should be configured. For example, after provisioning the Ubuntu server from the previous paragraph, we can automate the task of installing the Java Runtime Environment (JRE) 17 and opening the ports 8080 and 8443 from the firewall. Configuration as code applies to application configuration as well.

By automating all the infrastructure provisioning and configuration management tasks, we can avoid unstable, unreliable *snowflake servers*. When each server is provisioned, managed, and configured manually, the result is a *snowflake* : a fragile, unique server that cannot be reproduced and is risky to change. Automation helps avoid snowflakes in favor of *phoenix servers*: all tasks acting on those servers are automated, every change can be tracked in source control, reducing risks, and each setup is reproducible. By taking this concept to its extreme, we achieve the so-called *immutable servers*, also mentioned by the CNCF in its cloud native definition as immutable infrastructure.

NOTE You might have heard the expression “pets vs. cattle” when comparing traditional snowflake infrastructure (requiring a lot of care and attention, like pets) and immutable infrastructure or containers (characterized by being disposable and replaceable, like cattle). I won’t use this expression in the book, but it’s sometimes used in discussions about this subject, so you should be aware of it.

After their initial provisioning and configuration, immutable servers are not changed: they are immutable. If any change is necessary, it’s defined as code and delivered. A new server is then provisioned and configured from the new code while the previous server is destroyed.

For example, you have two options if your current infrastructure consists of Ubuntu 20.04 servers and you want to upgrade to Ubuntu 22.04. The first option is to define the upgrade via code and run the automation scripts to